

What is Excel?

→ Microsoft Excel is a spreadsheet software used to store, organize, analyze, and visualize data. Excel turns raw data into meaningful information and insights.

Purpose of Excel?

- • Clean Raw data
- Build interactive dashboards
- Visualize data trends
- Manage structured information

Types of data in excel?

- • Text :- names, categories
- Numbers :- sales, quantity
- Dates :- time based analysis
- Boolean :- True (1), False (0)
- Geographical data :- longitude, latitude.

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Excel feature for Data analytics

- • Data extraction
- Data cleaning
 - > Remove duplicate
 - > handling the missing values
 - > Removes the irrelevant data and adding necessary data
 - > Text Columns, Find and replace

• Formulas and functions

- > Basic :- Sum, average, Count
- > logical :- if, and, or, not
- > Lookup :- lookup, xlookup, vlookup, hlookup
- > Text :- Concat, upper, lower.

• Data Visualization

- > Charts :- Bar, Col, pie... etc
- > Conditional formatting
- > Sparklines

• Pivot tables

- > Summarize table
- > Group and filter data
- > Generate insights quickly

• Sorting and Filtering

• Data Validation (Extracting the proper data)

Advantages of Excel

- Beginner friendly
- Fast for small to medium dataset
- Widely used in Companies
- Integrates with tools like powerbi

Disadvantages of Excel

- Not suitable for large datasets
- Limited automation compared to coding, too.
- Manual errors are possible

Functions

Set of instruction / set of code is already pre-defined is called functions

There are 3 main Components of functions

- (i) Function name
- (ii) Number of inputs
- (iii) Return value

Lower Case

The lower function Converts all letters in a text string to lower letters.

Syntax :- =LOWER (text)

Eg :- =LOWER ('ArUn')

ArUn

Upper Case

The upper function Converts all letters in a text string to uppercase letters

Syntax :- =UPPER (text)

Eg :- =UPPER ('ArUn')

ARUN.

Proper Case

The proper function Converts the first letter of each word to uppercase and the remaining letters to lowercase

Syntax :- =PROPER (text)

Eg :- =PROPER ('ArUn')

ArUn.

Len

The len function counts the number of characters in a text string, including spaces.

Syntax :- = LEN (text)

Eg :- = LEN ("Data Analyst") → 12

Left function

The left function extracts a specified number of characters from the beginning (left side) of a text string.

Syntax :- = LEFT (text, num_chars)

Eg :- = LEFT ("Data Analyst", 4) → Data

Right function

The right function extracts a specified number of characters from the end (right side) of a text string.

Syntax :- = RIGHT (text, num_chars)

Eg :- = RIGHT ("Data Analyst", 7) → Analyst

Mid function

The mid function extracts a specific number of characters from the middle of a text string, starting from a specified position.

Syntax :- = MID (text, start_num, num_chars)

Eg :- = MID ("Data Analyst", 6, 7) → Analyst

Text extraction using left and len (First-half)

The left + len Combination is used to extract the first half of the characters from a text string.

Syntax :- = LEFT(text, LEN(text)/2)

Eg :- = LEFT(A1, LEN(A1)/2) Name:- Rahul

Result : RA

Second-half character extraction

The right + len Combination is used to extract the last half of a text string

Syntax :- = RIGHT(text, LEN(text)/2)

Eg :- If A1 Contains RAHUL.

= RIGHT(A1, LEN(A1)/2)

Result : UL

Round-up function

The roundup function rounds a number upward, away from zero, to a specified number of digits

Syntax :- = LEFT(text, ROUNDUP(LEN(text)/2, 0))

Eg :- If A1 Contains RAHUL.

= LEFT(A1, ROUNDUP(LEN(A1)/2, 0))

Result : RAH